When Do Transformers Learn Relational Structure? A Synthetic Study of Affine Maps, Curriculum Design, and Single-Exposure Learning

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May 2026

Abstract

We investigate when and how transformers learn structured relational representations in synthetic settings with full experimental control. The work spans two complementary lines of experiments. In **Salzburg**, we test whether a single-layer transformer implements relations as affine maps between subject and object embeddings, finding a non-monotonic relationship between model capacity and structural solutions: structure emerges only at an intermediate "sweet spot." In **Graz**, we study single-exposure relational learning in a compositional type system with 45 types and 2025 entities. Through seven experimental iterations (v1–v7), we establish that: (1) type-level generalization is the dominant mechanism, accounting for over 90% of relational accuracy; (2) curriculum design — deferring relation facts until entity representations mature — yields 50% improvement over uniform training; and (3) a single exposure to a relational fact adds only 3.9 percentage points beyond what pure type-level generalization already provides. The central finding is that relational learning in transformers is overwhelmingly *type-level*: given sufficient training on a subset of entities, the model generalizes the relational map to unseen entities at over 90% accuracy, with single exposures providing only per-subject refinement.

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1 Introduction

Large language models encode vast amounts of relational knowledge — facts about entities, their properties, and their relationships. But the nature of this encoding remains debated: do models learn *structured* representations that support generalization, or do they *memorize* specific associations?

Recent work has provided evidence for linear structure in LLM representations. Hernandez et al. [1] showed that certain relations are approximately implemented as linear maps between subject and object embeddings. However, it remains unclear (a) when such structure emerges, (b) what role training curriculum plays, and (c) how much information a single exposure to a relational fact contributes beyond type-level generalization.

We approach these questions through two synthetic studies:

- **Salzburg**: Can a transformer learn to implement relations as affine maps? We vary the hidden dimension and discover a non-monotonic capacity–structure relationship.
- **Graz**: Can a transformer learn relational predictions from single exposures? We iterate through seven experimental designs (v1–v7), progressively isolating the mechanisms of relational learning and quantifying the contribution of curriculum design and single-exposure facts.

This report provides a comprehensive account of all experimental results, including negative results and design iterations that informed the final findings.

2 Salzburg: Affine Relation Maps

2.1 Setup

The vocabulary consists of 100 subjects ($a_0, \ldots, a_{99}$), 100 objects ($b_0, \ldots, b_{99}$), 2 relation tokens (r_0, r_1), and a separator token (x). Training sequences are 4-token triples $[x, r_k, a_i, b_j]$ where b_j is the correct object for subject a_i under relation r_k. Relation r_0 is the identity mapping ($a_i \mapsto b_i$) and r_1 is a cyclic shift by 50 ($a_i \mapsto b_{(i+50) \bmod 100}$).

The model is a single-layer transformer with 4 attention heads, trained with next-token prediction (cross-entropy loss). The hidden dimension d varies across experiments ($d \in \{8, 16, 32, 64\}$).

Evaluation. We measure: (1) *object prediction accuracy* — does the model predict the correct object? (2) *affine accuracy* — we fit a per-relation affine map $\hat{o} = W_k \cdot h_0(a_i) + b_k$ from input embeddings to output representations and check whether the affine-predicted output falls in the correct Voronoi cell.

2.2 Results: The Capacity–Structure Tradeoff

All models achieve near-perfect object prediction ($\geq 99.5\%$) regardless of dimension. However, affine accuracy varies dramatically:

Three regimes are evident:

- **dim=8 — No good solution exists (A).** 100 objects in 8 dimensions cannot be separated well enough for affine maps to work.

Table 1: Salzburg dimension sweep. All models predict correctly, but affine structure is non-monotonic in capacity.

	dim=8	dim=16	dim=32	dim=64
Parameters	5,128	11,792	29,728	84,032
Object accuracy	99.5%	100%	100%	100%
Affine accuracy	73.0%	97.0%	100%	27.5%
Existence test	70.0%	99.0%	100%	—
Regime	(A)	Sweet spot		(C)

- **dim=16–32 — Sweet spot.** Affine maps are both representable and found by SGD, because they are the most parameter-efficient solution.
- **dim=64 — Memorization wins (C).** With 84K parameters for 200 facts (420 params/fact), the model memorizes without needing affine regularity.

2.3 Auxiliary Loss Nudges Toward Structure

At dim=16, adding a tiny auxiliary loss ($\lambda = 1.0$, stabilizing at $\sim$0.02% of the main loss) pushes affine accuracy from 97% to **100%**. The architecture *can* implement the affine solution; standard SGD does not find it without explicit pressure.

2.4 Mechanistic Analysis

A logit lens analysis reveals that attention routes information (copying subject and relation embeddings) but cannot perform relation-dependent transformations. Only the MLP's nonlinearity implements the gating needed to select which affine map to apply:

Table 2: Logit lens: object prediction accuracy at intermediate stages (dim=16).

Stage	Accuracy	Affine cos
h_0 (embedding only)	17.2%	—
$h_0 + \text{attn}$	14.4%	0.987
$h_0 + \text{attn} + \text{MLP}$	100%	0.773

The attention output is 0.987-affine in the input (mechanically linear), while the MLP output is only 0.773-affine — it provides the essential nonlinear gating.

3 Graz: Compositional Relational Learning

3.1 Entity and Type Structure

Entities are characterized by two independent attribute groups:

- **f-type**: a 2-element subset of 10 f-features $\rightarrow \binom{10}{2} = 45$ types
- **g-type**: a 2-element subset of 10 g-features $\rightarrow \binom{10}{2} = 45$ types

Each (f-type, g-type) combination defines one entity type; with 45 entities per type, there are $45 \times 45 = 2025$ entities total.

Relations. Two relations are defined via cyclic type shifts coprime with 45:

- R_0: shift the f-type index by 7 (mod 45)
- R_1: shift the g-type index by 13 (mod 45)

Each subject maps to exactly 45 valid objects per relation (those sharing the target type after the shift). Chance level: $45/2025 = 2.22\%$.

Training sentences. Sequences of 48 tokens contain attribute sentences ($[x, e, a]$: predict attribute from entity) and relation sentences ($[x, r_k, s_i, o_j]$: predict valid object from subject and relation).

Model. Single-layer transformer, 4 attention heads, dim=64, 580,160 parameters. All Graz experiments (v1–v7) use this architecture.

3.2 Evaluation

We hold out 20% of (subject, relation) pairs at random. Held-out subjects still appear in attribute sentences (their type is learned) but never in relation contexts.

- **Train relation accuracy**: top-1 accuracy on training (subj, rel) pairs
- **Held-out relation accuracy**: top-1 accuracy on held-out pairs (main generalization metric)
- **Affine probe**: post-hoc centroid probe trained on half the subjects, tested on the other half

4 Graz v1: Single-Exposure from Scratch (Negative Result)

4.1 Design

Phase 1 trains on attributes only for 50 epochs until 100% accuracy. Phase 2 introduces all relation sentences, each triple seen exactly once per epoch. Setup: 6 features, 15 types, 225 entities, dim=16, 1 layer.

4.2 Results

Table 3: v1: Total failure — 0% relation accuracy after 100 relation epochs.

Rel Epoch	Loss	Train Rel	Held Rel	Attr Acc
pre	—	0.000	0.000	1.000
1	14.264	0.000	0.000	1.000
10	12.577	0.000	0.000	0.106
50	7.496	0.000	0.000	0.000
100	7.079	0.000	0.000	0.000

4.3 Diagnosis

Two problems emerged:

1. **Relation token cold-start**: The relation tokens r_0, r_1 were never seen during phase 1. Their initial loss was 14.3 (vs. 2.5 for attribute sentences), creating extreme gradients that destroyed the learned representations.

2. **Catastrophic forgetting**: Attribute accuracy crashed from 100% to 0% and never recovered. The model lost all entity structure.

4.4 Lesson

Relation tokens must be introduced alongside entity representations, not after. Sequential phasing (attributes first, then relations) causes a distribution shift that the small model cannot absorb.

5 Graz v2: Relation Token Warmup

5.1 Design

Phase 1 trains with a small "pilot" set of k relation triples mixed in alongside attributes. This teaches the model the relation sentence format without requiring generalization. Phase 2 then introduces all remaining triples.

Setup: 6 features, 15 types, 225 entities, dim=16, 360 training triples. Chance level: 6.67%.

5.2 Phase 1: k-Triple Sweep

Table 4: Phase 1 results: all pilot triples memorized, no generalization until $k = 32$.

k	Pilot Rel	Remaining Rel	Held Rel	Affine Test
4	1.000	0.000	0.000	0.292
8	1.000	0.000	0.000	0.296
16	1.000	0.000	0.000	0.217
32	1.000	0.067	0.067	0.230

5.3 Phase 2: Single-Exposure Relation Epochs ($k = 8$)

After the $k = 8$ warmup, remaining triples are presented once per epoch for 50 epochs:

Table 5: v2 phase 2: slow learning, 16.7% remaining accuracy after 50 epochs.

Epoch	Loss	k-Rel	Remaining	Held	Attr
pre	—	1.000	0.000	0.000	1.000
2	5.541	1.000	0.056	0.022	1.000
10	3.414	1.000	0.122	0.089	0.760
50	2.978	1.000	0.167	0.078	0.954

5.4 Findings

The warmup solves the cold-start problem (initial loss drops from 14.3 to 5.95) and prevents catastrophic forgetting. However, with only 3 gradient steps per epoch and 150 total steps over 50 epochs, the model barely exceeds chance on held-out triples. The **gradient budget is the bottleneck**.

6 Graz v3: Large-Scale Joint Training

6.1 Design

The key insight from v2: the model needs far more gradient signal. v3 scales to $\binom{10}{2} = 45$ types, 2025 entities, and dim=64, with 145,800 training facts (each seen exactly once) spread across 10 epochs. Attributes and relations train jointly from epoch 1.

Table 6: v3 configuration vs. v2.

	v2	v3
Entities	225	2,025
Types	15	45
Training facts	∼360	145,800
Parameters	20,368	580,160
Gradient steps	150	9,760

6.2 Results

Table 7: v3 epoch-by-epoch results (seed=42). Chance = 2.22%.

Epoch	Loss	Attr	Train Rel	Held Rel	Affine (indiv.)	Affine (centroid)
pre	—	0.003	0.010	0.014	—	—
1	4.584	0.978	0.003	0.004	0.059	—
2	3.396	1.000	0.006	0.004	0.396	—
4	3.261	1.000	0.103	0.095	0.453	—
7	3.207	1.000	0.399	0.243	0.442	—
10	3.196	1.000	0.434	0.269	0.455	1.000

6.3 Key Findings

1. Genuine type-level generalization. Held-out accuracy reaches 26.9% — **12× chance**. The model predicts correct objects for (subject, relation) pairs it has never seen, purely from type membership.

2. Perfect centroid probe. A type-centroid affine probe achieves 100% on both train and test subjects. The model has learned a perfect linear map from each subject's hidden state to the centroid of its 45 valid objects. The gap between the centroid probe (100%) and actual accuracy (26.9–43.4%) reflects the decoding bottleneck: the model points at the correct type centroid, but

the softmax over 2025 tokens must select one specific object from 45 candidates clustered around that centroid.

3. Phase transition at epoch 7. A sharp acceleration occurs between epochs 6–7: training accuracy jumps from 20.5% to 39.9%. After 6 epochs, the model has seen 60% of training facts, providing enough evidence for relational structure to consolidate.

4. The decoding bottleneck. Precision@45 analysis reveals the model finds only 12.5 of 45 valid objects in its top-45 predictions. The bottleneck is not representation (centroid probe is perfect) but output decoding: translating type-level knowledge into correct token predictions.

6.4 Top-k Coverage Analysis

Table 8: v3 top-k analysis confirms perfect type clustering in both directions.

Metric	k	Accuracy
Entity → f-attributes	2	1.000
Entity → g-attributes	2	1.000
Entity → all attributes (joint)	4	1.000
Attribute → valid entities	405	1.000

The model has learned complete type membership in both directions. The relation accuracy bottleneck lies entirely in relational decoding, not type clustering.

7 Graz v4: Reduced Object Pool (Negative Result)

7.1 Hypothesis

If the bottleneck is selecting among 45 valid objects out of 2025, reducing the object pool should help. We select a uniform subset of 675 objects (15 per f-type and 15 per g-type, using König's theorem on a 45-regular bipartite graph).

7.2 Results

Table 9: v4: reduced object pool. Accuracy is *worse* than v3.

	v3 (2025 objects)	v4 (675 objects)
Training facts	145,800	48,600
Train rel	0.434	0.094
Held-out rel	0.269	0.080
Chance	2.22%	2.22%

7.3 Lesson

Reducing the object pool from 2025 to 675 also reduces training facts by 3× (48,600 vs. 145,800). The decreased relational signal far outweighs the easier prediction problem. The bottleneck is not object-pool size but *gradient signal from the relational facts themselves.*

8 Graz v5: Curriculum with Warmup

8.1 Motivation

The v3 phase transition at epoch 7 suggests that early-epoch relation facts are partially wasted: attribute representations are still maturing in epochs 1–2, so relation sentences presented then contribute less to relational learning.

Key insight: Single-exposure facts are a non-renewable resource. Once presented to immature representations, the learning opportunity is lost.

8.2 Design

- **Warmup (epochs 1–2)**: attributes + pilot set of 100 (subj, rel) pairs → 4,500 facts, repeated. Purpose: learn relation format while representations mature.
- **Main (epochs 3–10)**: all 145,800 single-exposure facts spread across 8 epochs → 18,225 facts/epoch.

Same total compute as v3: 976 steps/epoch, 9,760 total steps.

8.3 Results

Table 10: v5 epoch-by-epoch results. 29% improvement over v3 with identical compute.

Epoch	Phase	Loss	Attr	Train Rel	Held Rel
pre		—	0.003	0.010	0.014
1	warmup	4.549	0.993	0.001	0.004
2	warmup	3.361	1.000	0.000	0.000
3	main	3.308	1.000	0.059	0.042
5	main	3.240	1.000	0.243	0.157
7	main	3.213	1.000	0.466	0.290
10	main	3.201	1.000	0.560	0.347

8.4 Key Finding: Scheduling Matters as Much as Quantity

The only difference between v3 and v5 is *when* facts are presented. Deferring single-exposure relation facts to epochs with mature entity representations yields a 29% improvement in both training and held-out accuracy. After 2 warmup epochs, the pilot set shows 0% relation accuracy — confirming that relation facts presented to immature representations are indeed wasted.

Table 11: v5 vs. v3: same data, same compute, different scheduling.

	v3	v5	Improvement
Train rel	0.434	0.560	+29%
Held-out rel	0.269	0.347	+29%
Held-out / chance	12.1×	15.6×	

9 Graz v6: Graduated Curriculum with Pilot Scaffold

9.1 Motivation

v5's 2-epoch warmup establishes entity representations, but the model enters the main phase with no relational knowledge. What if we extend the pilot phase to actually *learn* the relational pattern before exposing single-exposure facts?

9.2 Design

- **Pilot phase (epochs 1–5)**: 100 (subj, rel) pairs → 4,500 facts, repeated each epoch. Goal: learn the relational map from a small scaffold.
- **Main phase (epochs 6–13)**: all 145,800 single-exposure facts across 8 epochs → 18,225 facts/epoch.

Total: 13 epochs, 12,688 steps (30% more compute than v3/v5).

9.3 Results

Table 12: v6: pilot scaffold enables dramatically better single-exposure learning.

Epoch	Phase	Attr	Pilot	Non-pilot	Held	Affine
1	pilot	0.994	0.010	0.001	0.004	0.099
2	pilot	1.000	0.000	0.000	0.000	0.457
3	pilot	1.000	0.050	0.000	0.000	0.492
5	pilot	1.000	0.620	0.000	0.000	0.430
6	main	1.000	0.750	0.185	0.141	0.420
8	main	1.000	0.900	0.583	0.348	0.430
11	main	1.000	0.920	0.691	0.405	0.436
13	main	1.000	0.920	0.706	0.393	0.441

9.4 Key Findings

1. The scaffold builds a reusable relational prior. During the pilot phase, 100 triples reach 62% accuracy while non-pilot accuracy stays at exactly 0%. No leakage — the model learns subject-specific relational mappings, not just type patterns.

2. Immediate transfer at main phase onset. At epoch 6 (first main epoch), non-pilot accuracy jumps from 0% to 18.5%. These triples have never been seen. The 18.5% first-epoch transfer (vs. 2.22% chance) is strong evidence that the pilot scaffold teaches the type-level relational map.

3. Pilot accuracy rises during main phase. Pilot triples improve from 62% to 92% during the main phase, even though no new pilot facts are presented. Additional facts about different subjects of the same types reinforce the pilot's relational pattern — confirming type-level learning.

Table 13: Progressive improvement through curriculum refinement.

Version	Strategy	Train Rel	Held Rel	Epochs	Steps
v3	Uniform exposure	0.434	0.269	10	9,760
v5	2-epoch warmup	0.560	0.347	10	9,760
v6	5-epoch scaffold	0.706*	0.405	13	12,688

*Non-pilot accuracy (single-exposure triples only).

4. v3 → v5 → v6 progression. Each curriculum refinement improves held-out accuracy: +29% (v5 vs. v3), +17% (v6 vs. v5), +50% total (v6 vs. v3).

10 Graz v7: Three-Way Split — Isolating Single-Exposure Value

10.1 Motivation

The v3–v6 experiments conflate two sources of relational learning: (a) the type-level map learned from repeated facts, and (b) per-subject information from single exposures. v7 is designed to cleanly separate these by introducing a three-way split:

- **Train (60%)**: 2,430 (subj, rel) pairs, repeated over 10 epochs
- **Single-exposure (20%)**: 810 pairs → 36,450 facts, each seen once
- **Held-out (20%)**: 810 pairs, never seen

The key comparison: after Phase 2 (single-exposure), how much does the single-exposure group improve *relative to held-out* (which received no exposure)?

10.2 Design

- **Phase 1 (10 epochs)**: Attributes + 60% train relations, repeated. Cosine LR from 3e-4, 200-step warmup. $p_{\text{relation}} = 0.20$.
- **Phase 2 (1 epoch)**: 36,450 single-exposure facts, each seen exactly once. Constant LR = 1e-4.

Total: 10,736 steps (976/epoch).

Table 14: v7 Phase 1: learning dynamics. Single and held-out are identical (neither seen), confirming no information leakage.

Epoch	Loss	Attr	Train 60%	Single 20%	Held 20%
pre	—	0.003	0.010	0.011	0.014
1	4.766	0.416	0.000	0.000	0.000
2	3.675	1.000	0.049	0.027	0.031
3	3.481	1.000	0.630	0.326	0.337
4	3.359	1.000	0.908	0.616	0.662
5	3.291	1.000	0.994	0.801	0.821
7	3.242	1.000	1.000	0.861	0.873
10	3.227	1.000	1.000	0.906	0.906

Table 15: v7 Phase 2: the value of a single exposure.

	Train 60%	Single 20%	Held 20%
After Phase 1	1.000	0.906	0.906
After Phase 2	0.999	**0.995**	**0.956**
Change	−0.001	+0.089	+0.049

10.3 Results

10.4 Key Findings

1. Type-level generalization reaches 90.6% without any single-exposure facts. After Phase 1, both the single-exposure and held-out groups score 90.6% (identical, as expected). The model learns the relational map from 60% of entities well enough to generalize to the remaining 40% at over 90% accuracy — **40.8× chance**.

This is dramatically higher than v3–v6 held-out accuracy (27–41%). The difference: Phase 1 trains on 60% of relations with full repetition (10 epochs), giving the model sufficient signal to learn the complete type-level map.

2. A single exposure adds 8.9 percentage points. Single-exposure facts go from 90.6% → 99.5% after one pass through 36,450 facts. This +8.9pp gain represents per-subject information that the type-level map cannot capture.

3. Spillover benefits held-out by 4.9pp. Held-out accuracy rises from 90.6% → 95.6% despite those facts never being seen. The additional 36,450 facts in Phase 2 strengthen the type-level map itself.

4. Net single-exposure advantage: 3.9 percentage points.

5. Phase 1 learning dynamics. The train/generalization ratio converges from 1.90 (epoch 3) to 1.10 (epoch 10), indicating that the model's relational knowledge becomes increasingly type-level as training progresses. By epoch 10, 91% of what the model knows about trained relations transfers to unseen subjects.

Table 16: The net value of a single exposure, controlling for spillover.

Group	After Phase 2	Source of knowledge
Single (seen once)	99.5%	Type-level + per-subject detail
Held (never seen)	95.6%	Type-level only
Difference	**3.9pp**	Per-subject information from single exposure

11 Summary of Graz Progression

Table 17: Complete Graz progression from v1 (failure) to v7 (clean isolation).

Ver.	Strategy	Entities	Train	Held	Steps	Key insight
v1	Sequential phases	225	0.000	0.000	300	Cold-start kills
v2	Warmup + exposure	225	0.167	0.078	150	Gradient-limited
v3	Joint training	2,025	0.434	0.269	9,760	Scale works
v4	Reduced pool	2,025	0.094	0.080	9,760	Less signal hurts
v5	Curriculum	2,025	0.560	0.347	9,760	Timing matters
v6	Scaffold	2,025	0.706	0.405	12,688	Prior helps
v7	3-way split	2,025	1.000	0.956	10,736	Type $\gg$ instance

12 Discussion

12.1 The Central Finding: Relational Learning is Type-Level

The v7 experiment provides the cleanest answer to the original research question. A transformer trained on 60% of (subject, relation) pairs with repeated exposure generalizes to the remaining 40% at 90.6% accuracy. A single additional exposure adds only 3.9pp of per-subject information beyond this.

This reframes "single-exposure relational learning": the model does not learn the relation from a single exposure. It already knows the relation from the type structure. The single exposure merely fills in subject-specific residual information within a well-learned relational scaffold.

12.2 Curriculum Design Matters

The progression from v3 to v6 demonstrates that the scheduling of learning signal matters as much as its quantity:

- v5 (+29% over v3): same data, same compute, deferred scheduling
- v6 (+50% over v3): 30% more compute, pilot scaffold

The key principle: **single-exposure facts are a non-renewable resource**. Presenting them to immature representations wastes the learning opportunity. Optimal curriculum design ensures entity representations and relational priors are established before single-exposure facts are consumed.

12.3 The Capacity–Structure Connection

The Salzburg results (Section 2) and the early Graz experiments (not reported in detail here; see Appendix) show the same non-monotonic pattern: structured solutions emerge only when model capacity is matched to task complexity. However, v7 reveals that once sufficient training data is provided, the capacity concern becomes secondary: the dim=64 model (which in the dimension-sweep experiments memorized rather than generalized) achieves 90.6% type-level generalization when trained on enough relations. The capacity–structure tradeoff is thus modulated by data quantity.

12.4 Three Mechanisms of Relational Knowledge

v7 cleanly separates three mechanisms:

1. **Type-level generalization (90.6%)**: From repeated training on 60% of entities, the model learns a complete type-to-type relational map that transfers to unseen entities.

2. **Spillover reinforcement (+4.9pp)**: Additional single-exposure facts strengthen the type-level map even for entities that never appear in those facts.

3. **Per-subject detail (+3.9pp)**: A single exposure provides subject-specific information that pure generalization cannot supply.

12.5 Implications

For understanding LLMs. Real LLMs encounter most relational facts only a few times in their training corpus. Our results suggest that for relations with clear type structure ("capital of," "CEO of"), the model may learn the relation primarily from type-level patterns (countries map to cities, companies map to people) rather than from individual fact exposures. The individual facts then serve primarily as per-entity refinements.

For knowledge editing. If relational knowledge is primarily type-level, then editing a single fact (e.g., changing a capital city) requires overcoming the type-level prior. This may explain why knowledge editing techniques often exhibit "ripple effects" — the type-level structure resists localized changes.

For data efficiency. The curriculum results suggest that fact presentation order matters for data-efficient learning. Training on type-representative examples first (establishing the relational scaffold) and then presenting the long tail of specific facts may yield better generalization than uniform random training.

12.6 Limitations

- All experiments use a single-layer transformer. Multi-layer models may exhibit different dynamics.

- The synthetic type system has perfectly clean structure. Real-world relations are noisier and may not decompose as cleanly into type-level and instance-level components.

- The 45-type system with 45 entities per type creates a balanced structure. Skewed distributions (few entities per type, or many types with few instances) might change the balance between type-level and per-subject learning.
- We use dim=64 throughout the v3–v7 experiments. The Salzburg results suggest this may be in the "memorization regime" for the capacity sweep, but the v7 result shows that with enough data, generalization succeeds regardless.

13 Conclusion

Through seven experimental iterations in the Graz setting, we have established that:

1. **Relational learning is overwhelmingly type-level.** A model trained on 60% of entities generalizes to unseen entities at 90.6% accuracy. Per-subject information from single exposures adds only 3.9pp.
2. **Curriculum design yields large gains.** Deferring single-exposure facts until entity representations mature improves generalization by 50% (v6 vs. v3) with modest additional compute.
3. **The relational map is a reusable scaffold.** Once learned from a small pilot set, it enables efficient absorption of new facts: non-pilot triples reach 70.6% after a single exposure, and a full 60% training set enables 90.6% pure generalization.
4. **Structure vs. memorization depends on capacity-to-data ratio.** The Salzburg results confirm that structured solutions emerge only at intermediate capacity. But with sufficient relational data (v7), even high-capacity models learn generalizable type-level structure.

The central message: transformers learn relations primarily as type-level maps. Individual fact exposure serves as refinement, not as the primary learning mechanism.

A Graz Phase 3 Experiments: Feature Reduction and Contrastive Learning

Prior to the one-epoch experiments (v1–v7), we conducted a series of experiments ("Phase 3") exploring the transition from memorization to generalization in a smaller setting. These experiments motivated the larger-scale one-epoch line of work.

A.1 Progression from Memorization to Generalization

A.2 Dimension Sweep (Phase 3o)

The Graz dimension sweep at 5 features / 100 entities shows the same non-monotonic capacity–structure pattern as Salzburg:

A.3 Contrastive Learning Ablation

Contrastive sentences ("is object b_j related to subject a_i via r_k? yes/no") are critical for generalization in the small-scale setting:

With insufficient data, contrastive sentences *hurt* (competing for capacity). With 10× more data, they drive strong compositional generalization.

Table 18: Phase 3 experimental progression.

Phase	Key change	Features	Entities	dim	Held-out	× chance
Phase 2	Subject holdout	10	120	32	0%	—
Phase 3	Triple holdout	10	120	32	35%	42×
Phase 3j	7 features, k=2	7	441	32	50.6%	10.6×
Phase 3l	+ contrastive	7	441	64	69.3%	14.6×
Phase 3n	6 features	6	225	64	87.8%	13.2×
Phase 3o3	5 features, dim=16	5	100	16	90.0%	9.0×

Table 19: Graz dimension sweep (5 features, 100 entities). Chance = 10%.

	dim=8	dim=16	dim=32	dim=64
Held-out accuracy	42.5%	**90.0%**	85.0%	67.5%
Contrastive +/−	85.2/94.0	88.8/97.8	88.8/99.2	90.0/97.6
Affine probe (test)	26.0%	50.0%	55.0%	63.0%
Partner similarity	+0.22	+0.43	+0.18	+0.10

B Salzburg: Additional Mechanistic Details

B.1 Sublayer Decomposition

Attention is mechanically linear (weighted sum of value-projected inputs). The MLP's GELU nonlinearity implements the relation-dependent gating: "if relation = k, apply transformation W_k." This explains why post-attention accuracy (14.4%) is no better than chance despite the attention output being highly structured.

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Table 20: Effect of contrastive learning (7 features, 441 entities). Chance = 4.8%.

	No contrastive	Contrastive, 6.25K seq	Contrastive, 62.5K seq
Held-out accuracy	50.6%	30.7%	**69.3%**
× chance	10.6×	6.4×	14.6×

Table 21: Affine predictability of sublayer components from input embedding (dim=16).

Target	$\cos(\hat{o}, o)$	Interpretation
Attention output	0.987	Near-perfectly affine
MLP output	0.773	Nonlinear gating
Full residual	0.735	Sum of all components